

Ayman Mahfuz

Software Engineer | Backend, Systems & Machine Learning Infrastructure

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EDUCATION

The University of Texas at Austin

Bachelor of Science in Computer Science

Austin, TX

Aug 2023 – May 2027

- **Concentration:** AI and ML — **GPA:** 3.5/4.0
- **Courses:** HPC, NLP, Generative Visual Computing, Energy-Efficient Computing

EXPERIENCE

Arm

May 2025 – Aug 2026

Platform Validation Intern – ML Infrastructure (Python, Bayesian Opt., PMU, Silicon)

- Built ML-guided CPU validation system cutting test executions **100x to under 1%** while preserving **99.8th-percentile failure coverage**; presented to SVP/director leadership.
- Architected modular **Python** search infrastructure with pluggable surrogate-model, acquisition-function, and kernel APIs, enabling new ML/validation strategies without core-pipeline changes.
- Applied **Bayesian optimization** and **PMU profiling** from tape-out through first- and post-silicon validation, exposing repeatable CPU/memory failures in ML workloads.

Texas Robotics

Jan 2025 – July 2026

Undergraduate Research Assistant – Reinforcement Learning (C++, Python, Multi-Agent RL, LLMs, GPU, CI)

- Built **C++/Python** sandboxes and hierarchical multi-agent **RL** environments, deploying trained policies to NAO robots within a **400K+ LOC** soccer system.
- Profiled and optimized **GPU utilization** and training pipelines, cutting iteration time **2–3x** and scaling experiments to **5M+ episodes**.

UT Austin – Center for Media Engagement

Aug 2023 – May 2025

Software Engineer, Undergraduate Research Assistant (BigQuery, Selenium, REST, JavaScript, PyTorch, BERT)

- Built Selenium/REST ingestion pipelines feeding **250M+ BigQuery** rows from **50M+ articles** and **70M+ comments** for large-scale ML research.
- Fine-tuned **BERT** classifiers for clickbait, entity, sentiment, and story identification, achieving **99% accuracy** through validation and error analysis.
- Developed **JavaScript** behavioral-research platform with round-robin game assignment, event deduplication, and structured logging for **1,000+ participants**.

UT Austin – Oden Institute

Aug 2023 – Jan 2025

Undergraduate Research Assistant – Scientific ML Infrastructure (PyTorch, Apptainer, SLURM)

- Engineered Apptainer/SLURM training pipeline on TACC H100s for 5-fold benchmarking of CNNs and transformers across **1,000+ MRI scans**.
- Benchmarked architectures, achieving **12% higher Dice**; separately resolved GPU-memory, I/O, and mixed-precision bottlenecks to stabilize PyTorch training.

PROJECTS

HelmPM (helmpm.app) – LLM-Native PM Platform | React, FastAPI, PostgreSQL, OpenAI/Anthropic

- Built multi-tenant React/FastAPI/PostgreSQL platform managing **thousands of tickets across 4 organizations**, with document/code retrieval and automated Linear/Jira sync.
- Engineered hybrid semantic/keyword retrieval and Merkle-tree indexing, enabling exact-ID search while re-embedding only changed documents.

253M-Parameter Transformer and RLHF Training Stack | PyTorch, HPC

- Built and pretrained **253M-parameter** PyTorch transformer on **600M tokens** using RoPE, RMSNorm, SwiGLU, and attention sinks.
- Engineered training stack with mixed precision, gradient accumulation, checkpoint recovery, evaluation, SFT, and DPO for reproducible multi-session HPC experiments.

LLM & Diffusion Energy Benchmark | Python, PyTorch, vLLM, SLURM, GH200

- Built Python/vLLM/SLURM energy benchmark using NVML and Zeus, running **577 tests across 55 configurations and 9 models** on GH200 GPUs.
- Measured **24–32% LLM energy reductions** from FP8/INT8 but zero diffusion savings; validated with Welch's t-tests/Bonferroni correction and explained via roofline analysis.

SKILLS

Systems/Performance: Python, C++, profiling, SLURM, Docker, Linux, vLLM, ZeusMonitor/NVML, TACC/HPC

ML/AI: PyTorch, Transformers, LoRA/PEFT, LLMs, Reinforcement Learning (PPO, MAPPO), RLHF/DPO

Backend/Web/Data: TypeScript, React, FastAPI, Flask, PostgreSQL, BigQuery, Pandas, NumPy, REST APIs, AWS, GCP

AWARDS & ACTIVITIES

RoboCup 2025 SPL – 3rd Place | Arm Patent-Filing Approval | Hack MSA – 2nd Place | UT AI Health Presenter